

# ReWIND: Psychoeducation Game Leveraging Cognitive Behavioral Therapy (CBT) to Enhance Emotion Control for Generalized Anxiety Disorder

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## ABSTRACT

ReWIND is a role-playing game (RPG) designed to enhance the emotion control of patients with generalized anxiety disorder (GAD) over the feeling of excessive worry and fear by integrating cognitive behavioral therapy (CBT) in a serious game. The goal of the game is to allow players to virtually encounter different anxiety-causing situations and provide constructive measures to deal with negative feelings following the ABC-model of CBT (antecedent, belief, consequence) along with disputation and new effect. The storyline's foundation focuses on four emotion regulation strategies common in GAD: catastrophizing, rumination, denial, and lack of refocus on planning. Each strategy consists of three scenarios that simulate real-life occurrences where GAD patients might find them hard to overcome. CBT elements are integrated with the diegetic components implemented in the game to provide psychoeducation in a fun way, which can be used to complement counseling.

## CCS CONCEPTS

• **Human-centered computing > Human computer interaction (HCI); • Human-centered computing > Interaction design;**

## KEYWORDS

Role-playing game (RPG), generalized anxiety disorder (GAD), cognitive behavioral therapy (CBT), emotion control

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## 1 INTRODUCTION

ReWIND is a serious game created with CBT experts to complement therapy for generalized anxiety disorder (GAD). Generalized anxiety disorder (GAD) is a relatively common disorder with high morbidity rate categorized by persistent or overly concerned about

opinions of others and excessive worry on imaginary or unnecessary matters [1]. The excessive worry or fear causes considerable misery on that person and subsequently affects daily life performance. Normal anxiety is a realistic response to a specific occurrence and stays for as long as the occurrence lasts. In fact, anxiety is a normal occurrence which allows a person to stay alert and react to stress or potential threat effectively. GAD on the other hand happens unexpectedly in random for seemingly no reason. It is characterized by unrealistic fear of a situation that will unlikely happen and persists for a long time even after the occurrence has been resolved. A person who has GAD may find it rather difficult to control or manage the emotion as well as the action. Therefore, GAD is a severe anxiety that requires treatment on learning how to control emotion in an effective way.

Cognitive behavioral therapy (CBT) is among the many solutions that can be used to treat GAD. CBT is a common goal-oriented psychotherapy treatment that tackles problem-solving through hands-on and practical approach [2]. Its main focus is to change the thoughts or behavioral patterns that are behind people's difficulties and subsequently improve their perceptions or feelings. CBT posits in any event, there will be antecedent (trigger), belief (interpretation) and consequence (reaction). In most cases, people feel anxious when they misinterpret the event and worry about it unnecessarily. At the same time, the use of medication alone to deal with GAD does not prevent it from recurring in the future. Thus, the key concept of CBT concentrates on the way we process and organize the information gathered, in order to allow a person to think in a more balanced and functional manner.

Normally, CBT is conducted by a psychotherapist in a limited number of sessions, whereby the psychotherapist would talk with the patients and help them to be aware of inaccurate negative thoughts so that the patient can respond or face the challenge in a more effective way. However, it is impossible for the psychotherapist to stay close to the patients at all time to provide continuous guidance especially during an unexpected episode of GAD. Thus, game approaches are integrated with CBT to allow a more flexible way to conduct the treatment, such as gamelets [3], game-based CBT (GB-CBT), and CBT-based serious game [4, 5]. GB-CBT is an integrative model that incorporates play therapy elements through developmentally appropriate games (DAGs) to administer treatment in a fun and engaging manner [6]. The GB-CBT model includes highly structured treatment sessions with rule-governed play and friendly competition as incentive for patients' participation. This allows a more fluid therapy process that can reduce defensiveness. CBT-based serious game on the other hand is a computerized cognitive behavioral therapy delivered in game style for treatment.

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It is fundamentally different from GB-CBT as CBT-based serious game delivers the therapy as a game as opposed to GB-CBT that integrates games during the therapy. However, existing studies have mainly focused only on depression, and a scarcity of research is observed for anxiety disorders [7]. In this project, a psychoeducation serious game leveraging CBT to enhance emotion control in GAD is proposed. As CBT in treating GAD includes a broad range of emotions, only four cognitive emotional regulation strategies are selected for the context, which are catastrophizing, rumination, denial, and lack of refocus on planning.

## 2 GOALS

The goals for the game, ReWIND include the following:

- Design interactive scenarios and storylines based on CBT for GAD to complement therapy.
- Provide the player with a learning environment that can provide opportunity for developing and improving on-the-spot decision-making skills for more effective emotion regulation in the onset of anxiety.
- Deliver psychoeducation in a fun and interactive way to help GAD patients be more self-aware of their cognitive distortions and discover strategies to turn negative thoughts into positive ones.

## 3 DESIGN

ReWIND is a single-player, CBT-based serious game in RPG style in which the player is reborn into the fantasy world as the pathfinder named Lucas to help people in overcoming life hardships. Patients or participants who are diagnosed with GAD symptoms will play the game as the main character to stimulate their imagination for different scenarios related to anxiety and think more critically through Lucas's perspective how the situations can be resolved. The game is designed to provide constructive measures for dealing with negative feelings in real life, such as learning on how to discharge emotion tension and so on to help players experience how changing their thoughts and behaviors in crises as they proceed with the story can lead to more positive outcomes. In addition, the game aims to improve the emotion control in keeping the feeling of excessive worry or fear at bay. The game is infused with CBT principles to show players the correct way to deal with various difficulties in life and prepare them for unknown situations that may occur in the future. Different aspects are combined together in order to achieve the desired outcome, such as the storyline, game mechanics, graphic design, and interaction design.

### 3.1 Storyline

Each storyline is anchored on the ABCDE-model of CBT: activating event, belief (cognitive distortion), consequence [3], along with disputation and effect [8]. The storyline is divided into four different chapters based on emotion dysregulation commonly found in GAD patients, which are catastrophizing, rumination, denial and lack of refocus on planning [9, 10]. Each chapter consists of three scenarios simulating real-life occurrences where GAD patients might find hard to overcome. Some elements of fantasy are infused into the storylines to fit the theme of the game and keep the game interesting to sustain the player's interest. The non-playing characters (NPCs)

in the game act as the quest givers or event triggers responsible for presenting the story in a diegetic manner. Meanwhile, players will have to assist the NPCs in resolving their problems. Table 1 shows two storylines from the catastrophizing chapter and the mapping of the storyline to the ABCDE-model.

### 3.2 Game Mechanics

There game mechanics used in the game keep the players continuously engaged in the gameplay to lower down their defense in internalizing the scenarios, including the quest system, inventory system, tutorial system, and stat system. Screenshots of these systems are shown in Figure 1.

**Quest system:** Players can go around and talk to quest givers and accept new quest, in which they will play the role as a pathfinder to assist the quest giver in resolving the problem. There are four different types of quest: maze-solving (finding a way through the maze to a key figure that is required to help the NPC change a cognitive distortion), searching (search for an item to reveal the cognitive distortion), item gathering (collect items requested by the NPC) and monster hunting (kill monsters). While the quest system connects the players to the core element that deliver the CBT content, it also adds some entertainment elements to the gameplay and enriches the storylines. Once players accept the quest, the quest giver will follow players around until the quest is completed. Players can also open the quest panel and check the current progress as well as the current task so players can keep track on what to do next.

**Inventory system:** Players can pick up items and store in the inventory. One of the quest types is item gathering which requires players to explore the game world and collect a fixed number of items to clear the quest.

**Stat system:** Upon quest completion, players will gain tokens that can be used to upgrade character stat. Stats available for the players include Positive Point (PP), Strength (STR) and Defense (DEF). The higher the character stats, the more advantages players have along the progress. For example, players can defeat stronger monsters with ease, lower damage taken from monsters, and make the screen gets darker at a slower pace. Players can also find tokens from treasure chests scattered around the game world to attract players to explore the game. The player's PP will reduce over time or when he or she is getting attacked by the monsters. The PP will affect the real-time environment so constant monitoring is required.

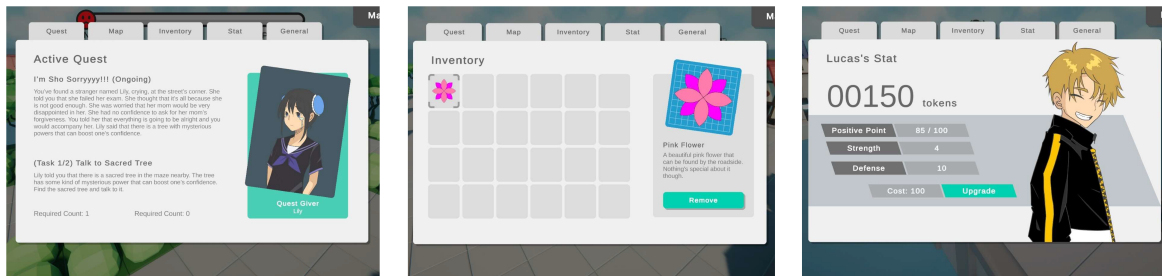
### 3.3 Graphic Design

Graphic design is leveraged to ensure that important messages and takeaways are communicated in more impactful ways to the players. The font focuses on human-centered approach whereby effects like text color and bolding are applied to emphasize key text to the player. This way, the player can easily catch the key takeaways in each storyline. An example of text font and color differences in the dialogue is shown in Figure 2

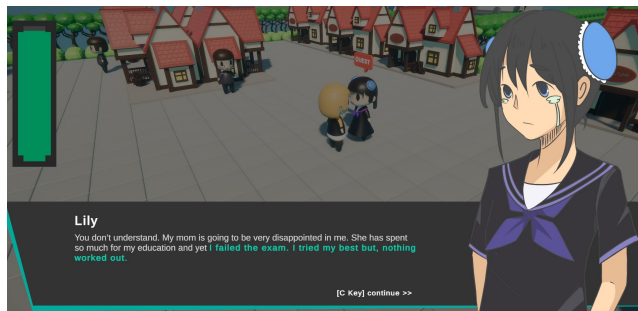
One innovative design in this game is the player will have a positive point bar (PP bar) that affects the color saturation of the game. The color saturation changes dynamically based on the condition of the player. For instance, if the player has lower PP then the color saturation in the game world will be lower; if the player has full PP, the color saturation is set to normal. Studies show that saturation is

**Table 1: Mapping the storylines to the ABCDE-model of CBT**

| Scenario | Antecedent                                             | Belief                                                                           | Consequence                                                                                                                                                                                                                        | Disputation                                                                                                                                                                           | Effect                                                                                                                                           |
|----------|--------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Lily failed her exam.                                  | Lily believes she is not good enough and her mother will be disappointed in her. | Lily cries at the street corner. She has no confidence to ask her mother for forgiveness. Lily said that there is a tree with mysterious powers that can boost one's confidence (she starts to question the cognitive distortion). | Lily questions her thoughts after hearing the tree. Lily tells herself that failing her exam does not mean she is a total failure. She braced herself and told her mother everything. | Lily realized things might not always be that bad. Lily gains back her confidence and promises to have more faith in herself.                    |
| 2        | Ben does not receive a reply from the person he likes. | Ben believes that no one in this world wants to gout with him.                   | Ben feels nervous and upset as he considers a future alone. He eventually considers looking for a technician that might give him a better solution. (he starts to question the cognitive distortion).                              | Ben questions his thought after hearing the technician and you out. He tells himself that he should stop magnifying the problem and take a step towards solving the problem.          | Ben realized things might not always be that bad. He gains back his confidence after getting a reply and promises to have more faith in himself. |



**Figure 1: In-game screenshots of quest system, inventory system, and stat system**



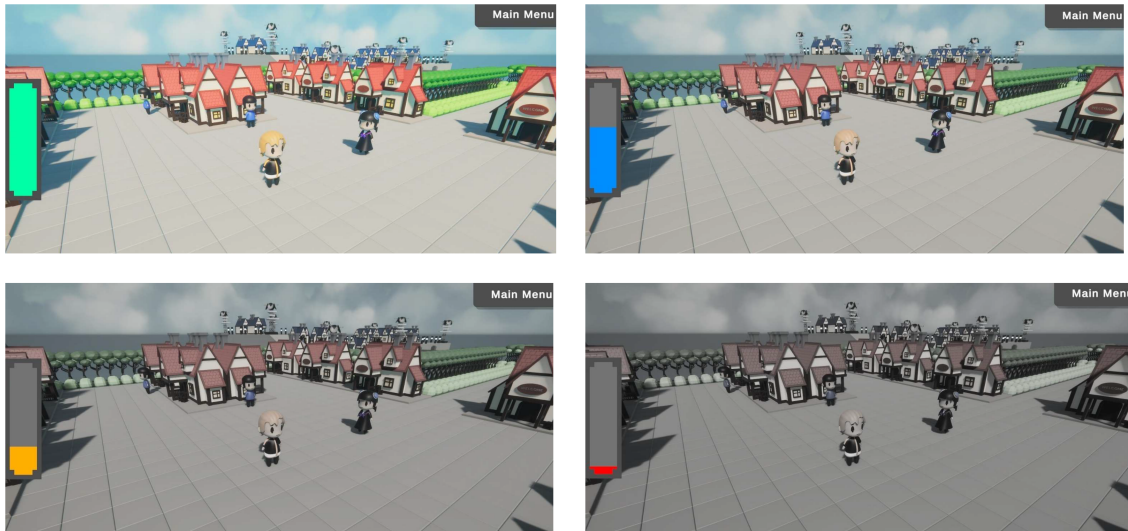
**Figure 2: In-game screenshot of different text color and bold for key text**

relative to the emotion of a person [11, 12], thus inspiring the idea of manipulating the color saturation to represent the emotional state of the player in the game. The player starts off with full PP and the game world is filled with bright colors. PP is reduced over time and, the player will view a darker screen if he or she cannot resolve the quest on time or find a way to replenish the PP. Dark and dull colors symbolize a negative emotional state when a cognitive distortion manifests through the game world and the only

way to turn the world bright again is to find positive outcomes in the storyline [12-14]. Figure 3 shows the color saturation difference based on the amount of PP of the player.

### 3.4 Interaction Design

To further improve the interaction between the player and the game, two crucial designs have been added to the game that links the player with the CBT elements, which are dialogue with decision making and emotion regulation progress update. The game attempts to educate the player to practice proper emotion regulation strategies when facing a scenario triggering anxiety by having correct mindsets to overcome their problems in a more rational and structural way. First, the decision- making has been added to the dialogue so that the player can have a more practical approach to learn the correct mindset when facing a certain problem. The decision-making element is suitable for creating a learning environment that can provide an opportunity for developing and improving on-the-spot decision making skill for more effective emotion regulation. It also makes the typical wordy dialogues more interactive to catch the player's attention. Dialogues with decision making becomes available after the player accepts the quest and are initiated by talking to the NPCs apart from the quest giver. If



**Figure 3: In-game screenshots of different color saturation based on Positive Point (PP)**

the player chooses the more appropriate answer, the player's emotion regulation progress increases. However, if a less appropriate answer is chosen, no progress is rewarded. The player may talk to the NPC again to retry the question. The question will be available until the appropriate answer is selected or upon quest completion. Through this dialogue design, I aim to provide the player with a learning environment for developing and improving on-the-spot decision-making skills for more effective emotion regulation in the onset of anxiety, which conforms with one of this project's goals. Figure 4 shows a screenshot of the dialogue with decision-making in ReWIND.

Second, the Emotivate Bar shown in Figure 5 is added to the game to represent the emotion regulation progress. It is located at the top of the screen and will only appear when the player accepts a quest. The Emotivate Bar shows the current emotion regulation progress of the quest giver. The emoji represents the current emotional state of the quest giver. The player is required to assist the quest giver in resolving the problem by changing the quest giver's cognitive distortions in order to reach the optimal ending.



**Figure 4: In-game screenshot of dialogue with decision making**

There are two ways for the player to gain progress for the bar: One, the more appropriate answer is chosen for dialogues with decision making; Two, the player clears a task of the quest (normally a quest will have 2 to 3 tasks). However, if the player only clears the tasks without talking to any NPC (not triggering any dialogue with decision making), then the maximum progress that the player can achieve is 70%. To obtain full progress, the player is required to explore the game and communicate with the NPCs. Dialogues with decision making will not appear before accepting the quest and after the quest is completed as the decision making is directed towards the quest giver encountering the specific problem in the context of the storyline. The player observes the whole storyline from Lucas's perspective (main character), thus gaining insight from a third person view.

Upon reaching 100% progress for the Emotivate Bar, the player is rewarded a bonus at the completion of the quest. This works as an incentive to encourage the player to explore and to communicate with other characters in the game to instill the importance of communication when a problem occurs. When the Emotivate Bar's progress is 30% or lower, the emoji will appear as a sad face. The emoji changes to a neutral face when the progress is between 31% to 65%. Lastly, a happy face emoji is shown when the emotion regulation progress reaches above 65%.

#### 4 DEVELOPMENT & TEST PLAN

The game development for this study is a solo effort that has been carried out for the past 12-week course of study. Weekly meetings with academic advisor were held to discuss progress, as well as carrying out interviews with psychology experts in CBT and GAD, who also served as the main reviewers for the project. Playtesting sessions were also carried out to gather feedback.

The submitted prototype is still in development stage at around 50% completion, including main system such as quest system, stat system, as well as two scenarios from the overall storyline. This is due to the limited manpower available whereby only one student



Figure 5: In-game screenshots of Emotivate Bar

is involved in all development aspects including the programming, design, illustration and so on. The prototype and the actual game will be available on PC platform with Windows operating system, and the game is developed using Unity game engine. Note that all the assets used in the game are prepared by the student without outsourcing.

## 5 EVALUATION PLAN

Once the game prototype is ready, the plan is to use a between-groups experimental design to evaluate the effect of playing ReWIND between participants with and without GAD. The evaluation includes participant recruitment and gameplay experiments.

**Participant recruitment:** Individuals with anxiety symptoms will be recruited to take part in the study to identify the effectiveness of ReWIND in improving emotion control for excessive worry or fear. Consent will first be obtained from the participants. Participants are required to fill in the symptom severity assessment form that is based on the GAD-7 scale [7] different group assignment.

**Gameplay Experiment:** Participant grouping can be divided into two categories: Group A consisting of participants showing no anxiety symptoms, and Group B consisting of participants showing anxiety symptoms. Lab experiments requiring participants to play the game for a fixed duration will be conducted. All participants are required to fill up the GADSS [15] and CERQ-short [16] questionnaires during pre-test (before playing the game) and post-test (after completing all 4 chapters of the game). We expect to observe a significant difference of anxiety symptom severity and cognitive emotion regulation between pre-test and post-test measures in Group B and no difference at all in Group A.

## 6 FUTURE WORK

Future work will focus on completing the development of ReWIND to include the remaining storylines with additional game mechanisms that can make the game more appealing and user-friendly. More playtesting and evaluation will be made alongside CBT experts and GAD patients to ensure CBT is implemented in an impactful and fun way. The upcoming updates include more quest types such as monster hunting and item gathering, and ensuring the CBT concepts are communicated clearly through the game.

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